Project 4

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SQL Query Generator

Natural Language to SQL with IBM Granite 3.3 2B

ABSTRACT

SQL Query Generation System is an intelligent platform that converts **natural language questions into SQL queries automatically**. The system helps users who do not have deep knowledge of SQL to easily interact with databases.

The system uses **Natural Language Processing (NLP)** and machine learning techniques to understand user questions and generate appropriate SQL queries. It analyzes the input sentence, identifies important keywords such as table names, columns, and conditions, and converts them into a valid SQL statement.

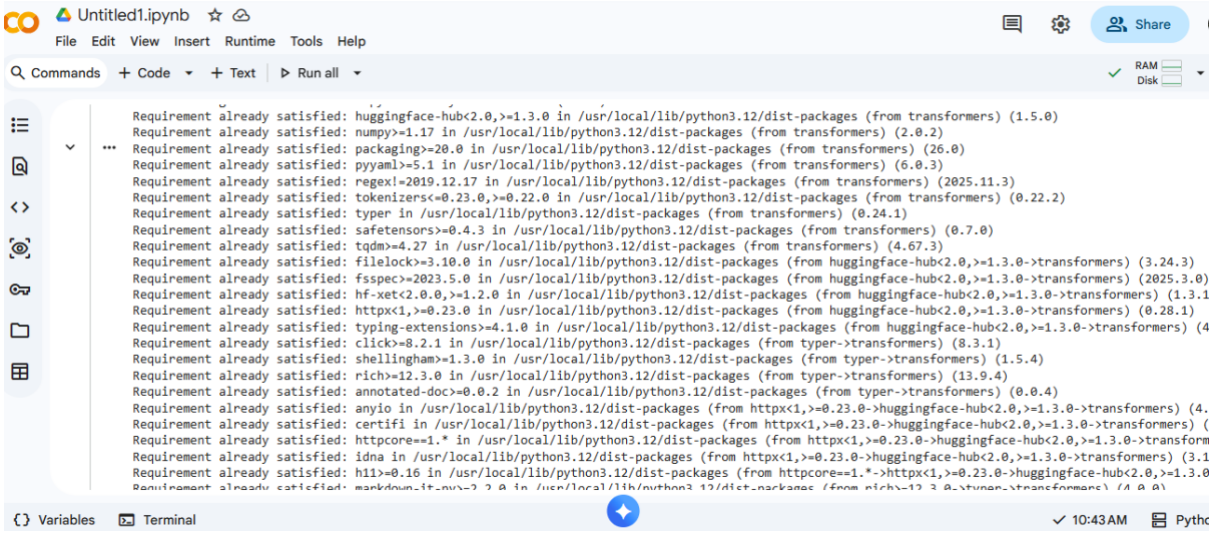
The platform provides a user-friendly interface where users can enter their questions in simple English. The system then generates the SQL query and executes it on the database to display the result.

This project reduces the difficulty of writing complex SQL queries and improves productivity for database users.

Step 01: Setup the Environment

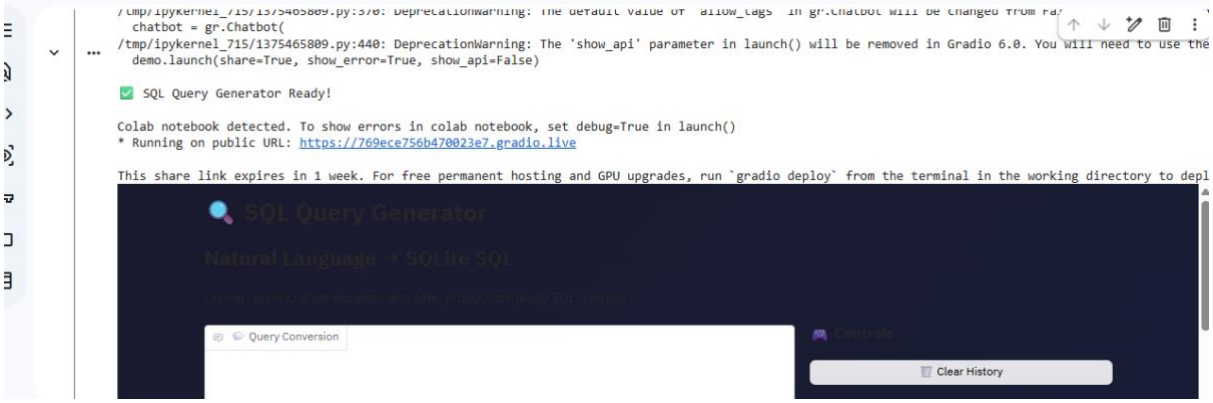
First, install and prepare the required tools for the project.
The programming language **Python** is used to develop the system. The database is created using **MySQL**. The code can be executed using **Google Colab** or a local Python environment.

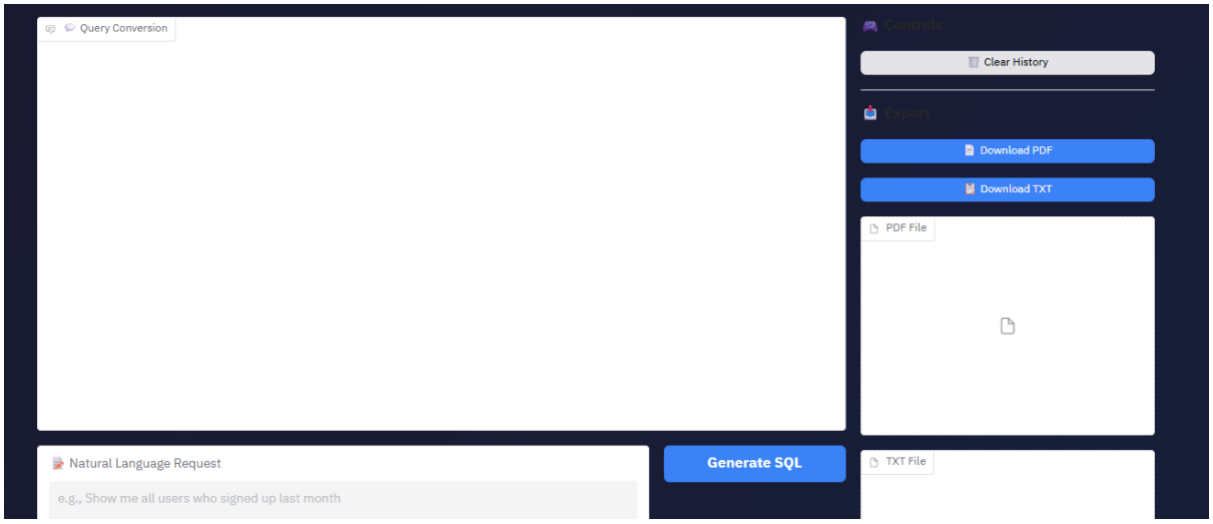
Required libraries such as NLP libraries and database connectors are installed to enable communication between the program and the database.



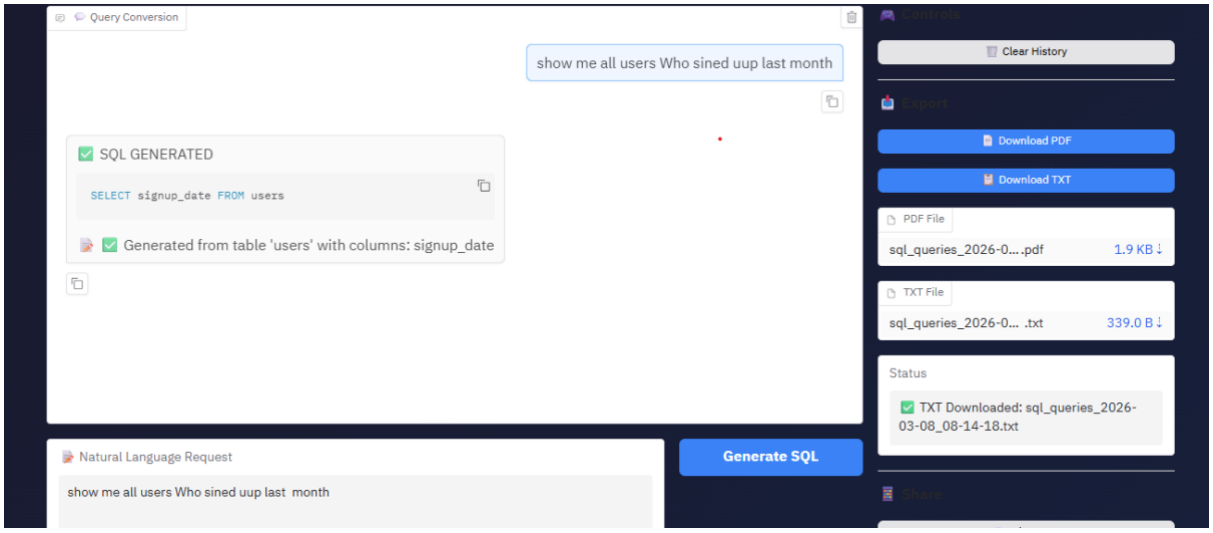
Step 02 :

After generating the link , by clicking that link it opens an environment to run our project





This is an interface to run the code to get code's document , while clicking the Generate Docstring it gives an output .



Here we can download the document in pdf format and aswellas txt format and the following screenshots contains the document we run in google and numPy format

```
SQL Query History Report
Generated: 2026-03-08 08:14:18.300748
============================================================

Query #1
User Request: show me all users Who sined uup last  month


Generated SQL: SELECT signup_date FROM users
Timestamp: 2026-03-08T08:14:07.729775
------------------------------------------------------------
```

These are the outputs for our project , This project helps us to generate document for our code in pdf and text formats, This is the final conclusion for this project.

THANKYOU